DATA VISUALISATION 2 · AUSTRALIAN MUSEUM COLLECTIONS

Australian Museum Collections

Australia's eight state and territory museums hold more than **64 million** objects and specimens — beetles pinned in 1890, shells dredged off the reef, fossils, stone tools, family heirlooms. What exactly is in those drawers, where did it come from, and how much of it can you actually see?

63.9 M
objects & specimens held by the seven museums that publish a total

6.1 M
specimen records you can search online, through the Atlas of Living Australia

≈ 9
online records for every 100 items in the stores

223,146
"type" specimens — the physical originals behind species names

By **Feng Qianjun** · FIT3179, Monash University , 21 October 2026

NO. 01 · THE MUSEUMS

Eight museums, founded over almost 200 years

Every state and territory runs its own museum. The oldest, the **Australian Museum** in Sydney, opened in 1827, only 39 years after the First Fleet. The youngest, the **Museum and Art Gallery of the Northern Territory**, opened in Darwin in 1981.

Size and age don't line up neatly. Sydney, Melbourne and Brisbane each look after about 15 million items or more, three times anything held elsewhere. Hobart's museum is the second oldest, yet it holds about 800,000.

Where the museums are

Circle area = objects and specimens held



Collection size, largest to smallest

Millions of objects and specimens, with founding year



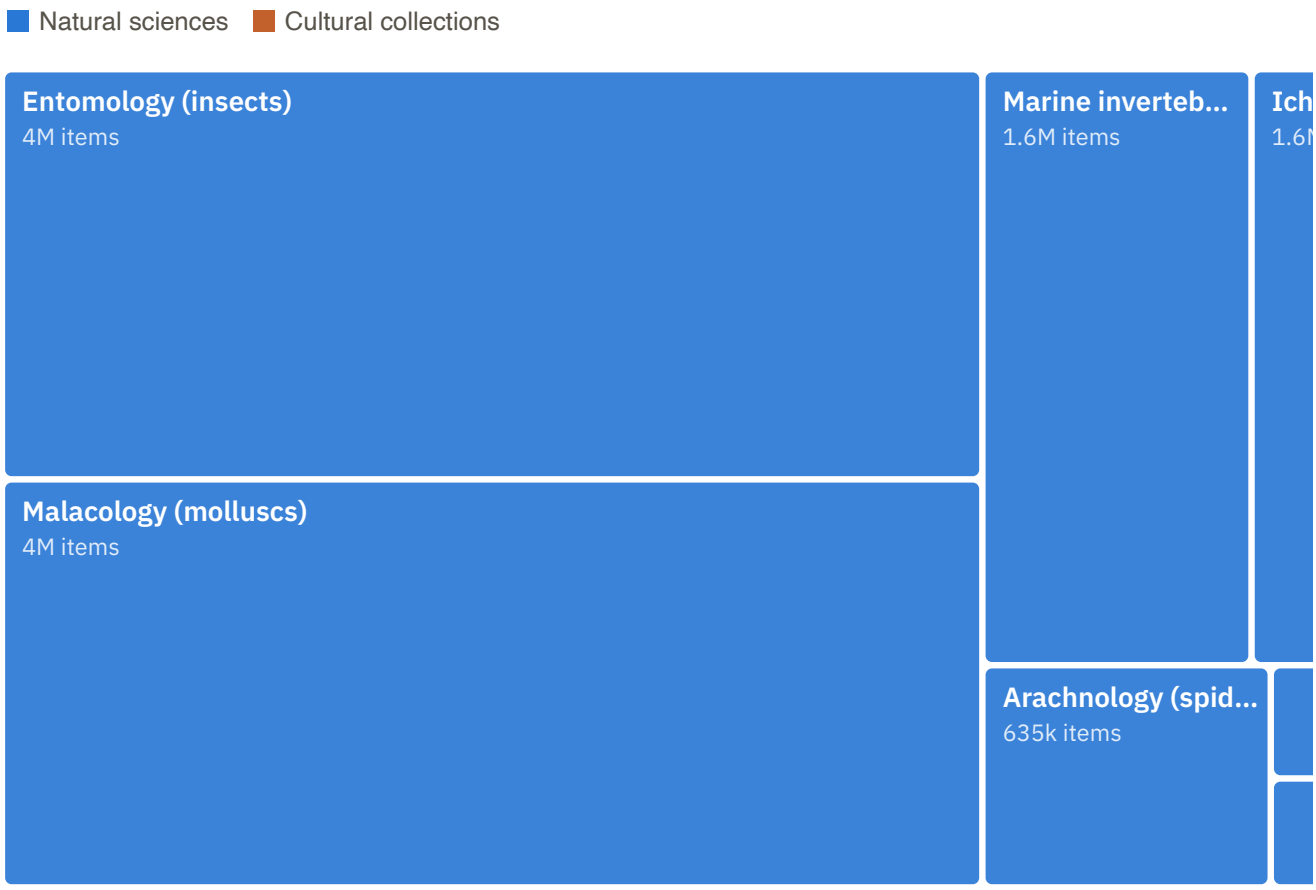
NO. 02 · WHAT'S IN THE COLLECTION

Most items are small animals, like insects and shells

Open the Australian Museum's stores and you mostly find **tiny animals.** Insects and molluscs make up about 4 million items each. Marine invertebrates and fish add another 1.6 million each. The cultural collections are smaller in number but no less important: around 800,000 First Nations archaeological artefacts, and about 105,000 First Nations, Pasifika and world-culture objects.

What the Australian Museum's departments hold

Rectangle area = number of items (approximate figures published by the museum)



The departments listed add up to about 13.3 million items. The museum's headline figure is "more than 22 million", and it does not break down the rest.

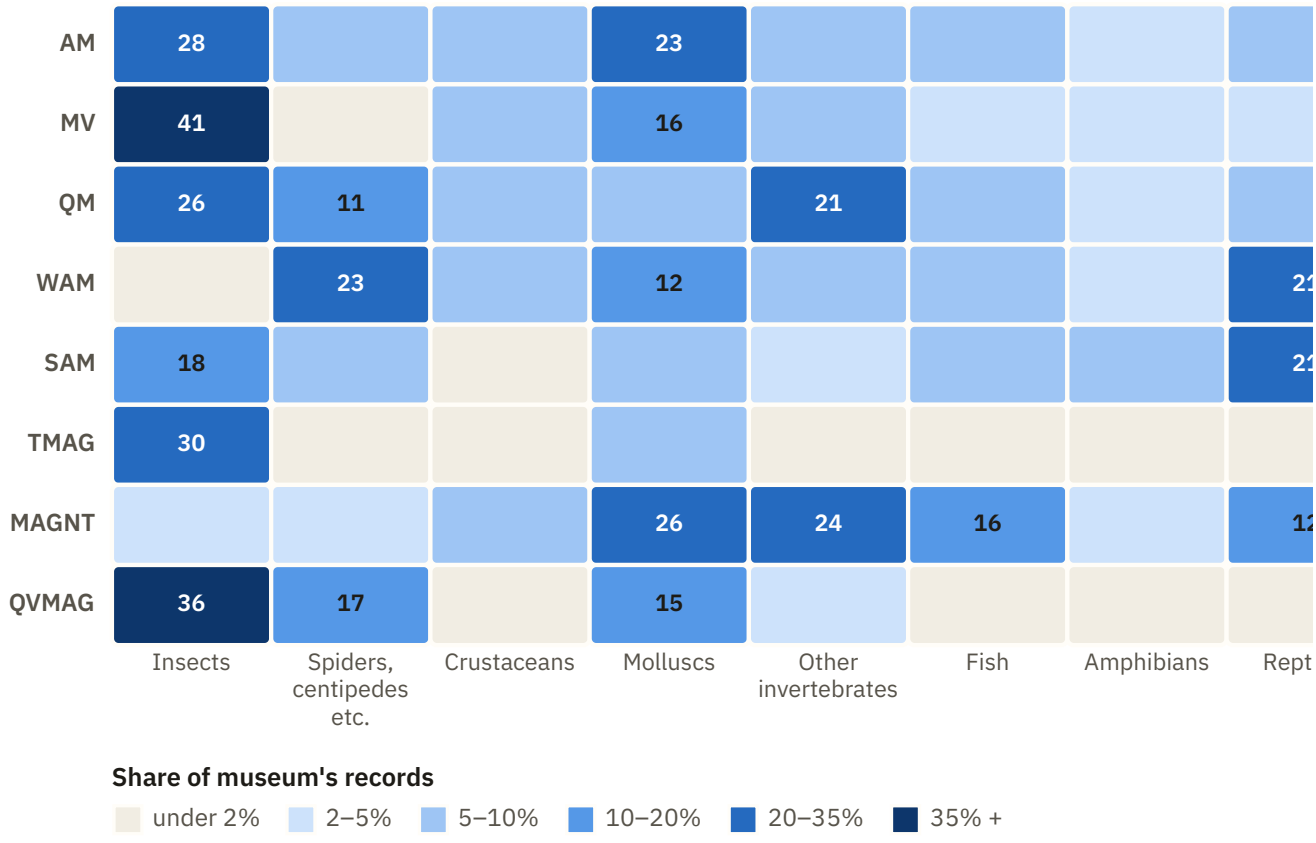
NO. 03 · WHAT EACH MUSEUM IS KNOWN FOR

Each museum collects different kinds of animals and plants

Natural history specimens have been digitised much faster than cultural objects, and they're shared on one national platform, the **Atlas of Living Australia (ALA)**. That means we can compare museums like for like. Each row below splits a museum's online records into groups of animals and plants. The darker a cell, the bigger that group's share.

What each museum's online specimen records are

Share of each museum's records (row sums to 100%)



Hobart is half botany. TMAG runs the Tasmanian Herbarium, so 51% of its records are plants and fungi.

Perth and Adelaide are reptile country. About one in five records at the WA and SA museums is a reptile. WA also leads on spiders and other arthropods (23%).

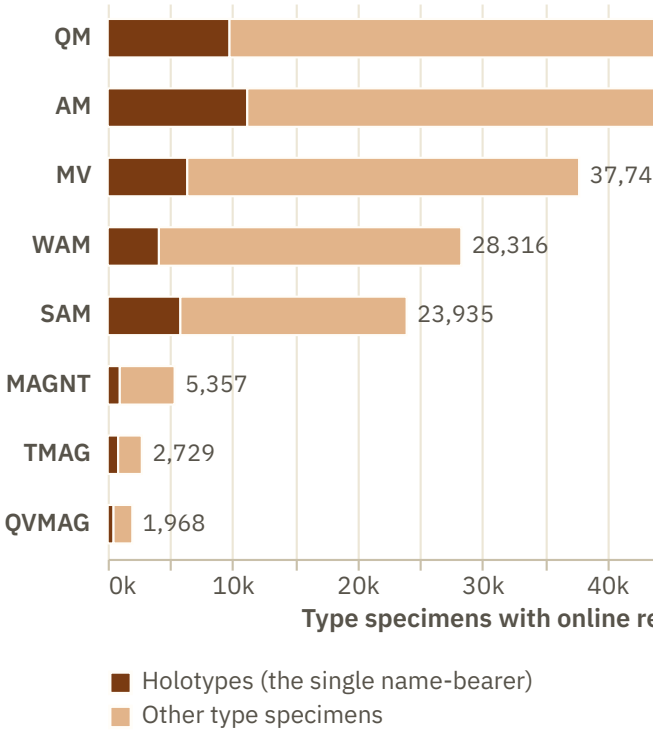
Melbourne is an insect

The name-bearers

Type specimens: the originals a species name is tied to

museum. 41% of Museums Victoria's records are insects.

A gap, not a fact. The WA Museum lists only 51 insects online. Its insect collection simply hasn't been shared yet.



When a scientist names a new species, one specimen, the **holotype**, becomes its permanent reference. Disputes about what a name means are settled by going back to that one animal in its drawer. Queensland and Sydney each hold about 60,000 type specimens, and Sydney alone holds over 11,000 holotypes.

NO. 04 · WHERE SPECIMENS COME FROM

Most museums collect mainly in their own state

Pick a museum. The map shows where its specimens were collected, out of all records with a known Australian state.

Where one museum's specimens come from

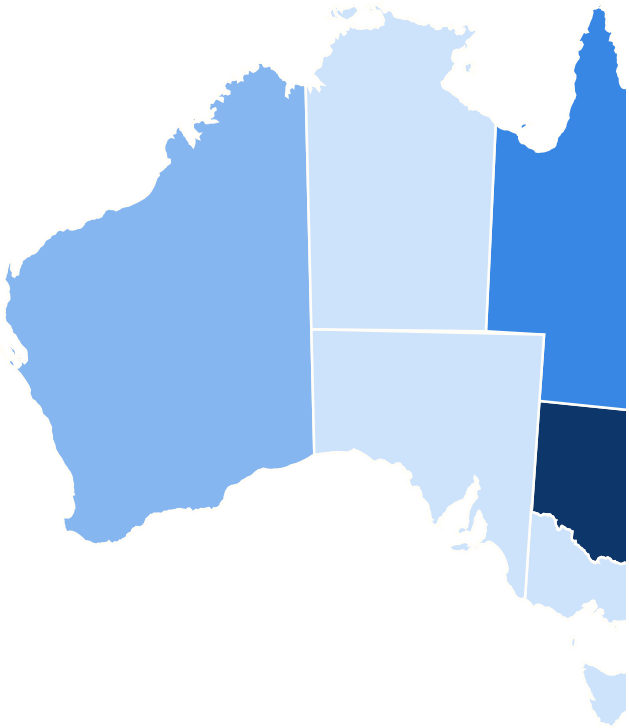
% of records with a known state

Most museums collect in their own backyard. Launceston, Perth, Hobart and Brisbane get **86–91%** of their located specimens from their home state. Sydney and Melbourne are the national collectors: almost half of their specimens come from other states or overseas. Queensland is Sydney's biggest "away" ground.

◆ marks the museum's home city. Use the menu above the map to switch museums.

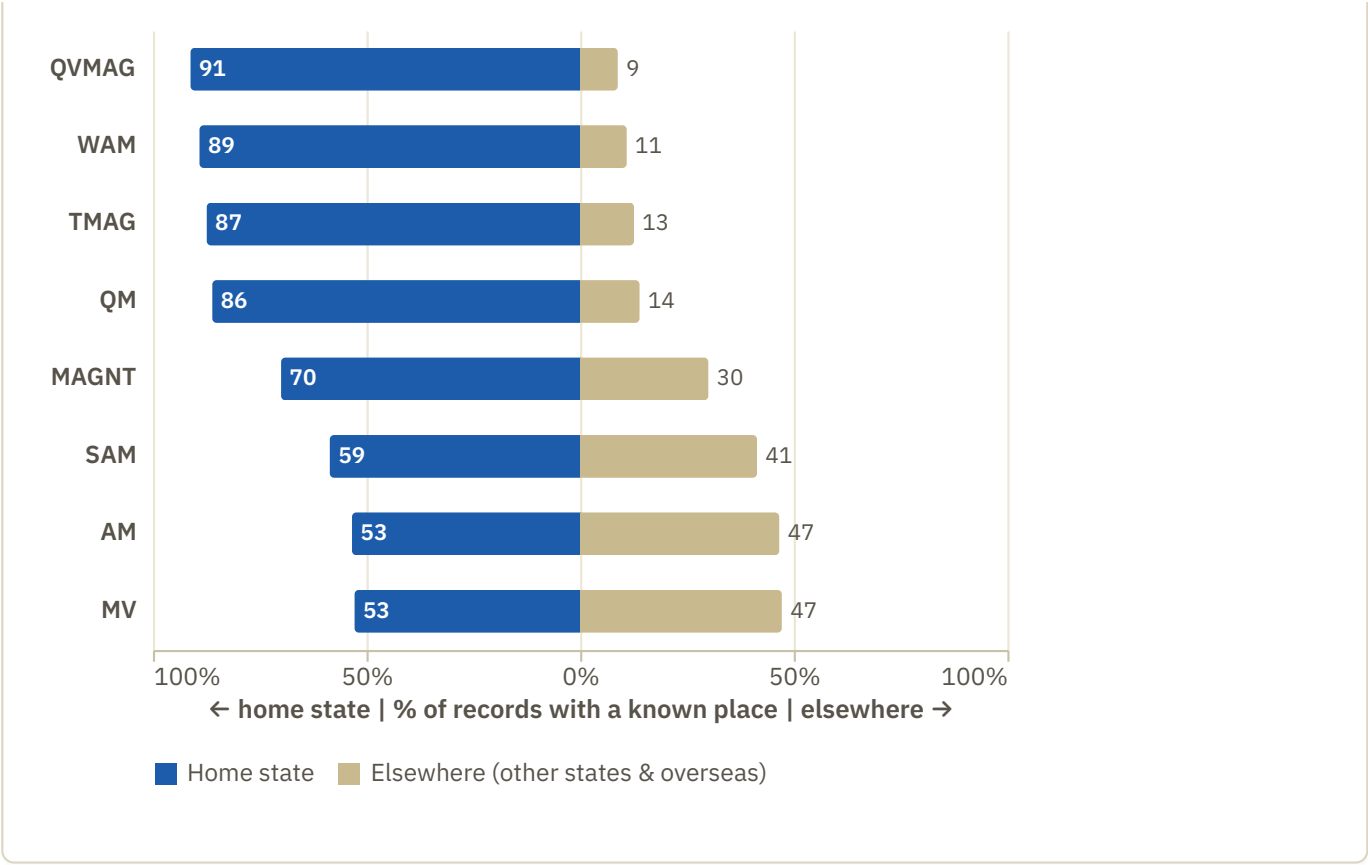
Choose a museum

Australian Museum



Home versus away

Share of records with a known place: home state (left) vs elsewhere (right)



NO. 05 · COLLECTING ACROSS AUSTRALIA

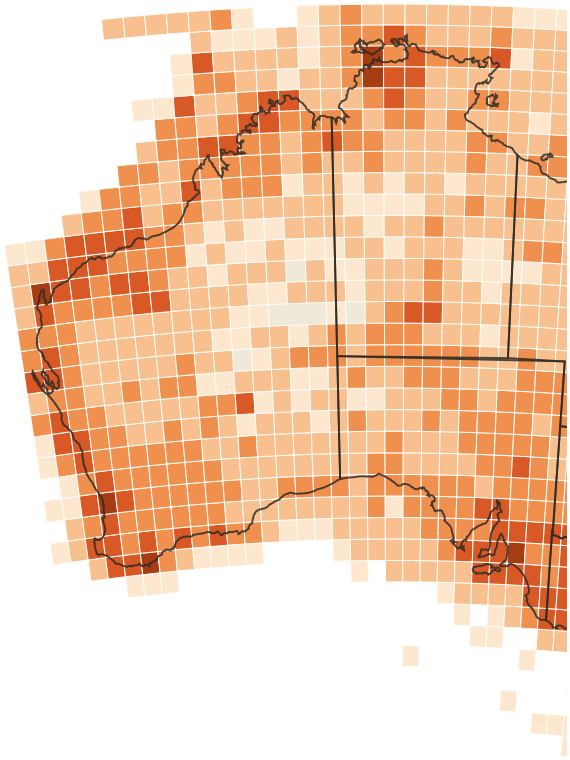
Where in Australia specimens have been collected

Split Australia into squares about 100 km wide and count the specimens collected in each. The left map shows the national picture of collecting effort: dense around capital cities, the Great Barrier Reef and Tasmania, and thin across the arid interior. A single square containing Sydney holds **233,425** records, more than the whole online collection of Launceston's museum.

The right map colours each square by the museum that holds the most specimens from it. The borders between museum "territories" closely follow state borders. Click a museum in the legend to see its territory.

How much has been collected, where

Specimen records per 1° × 1° square, all eight museums

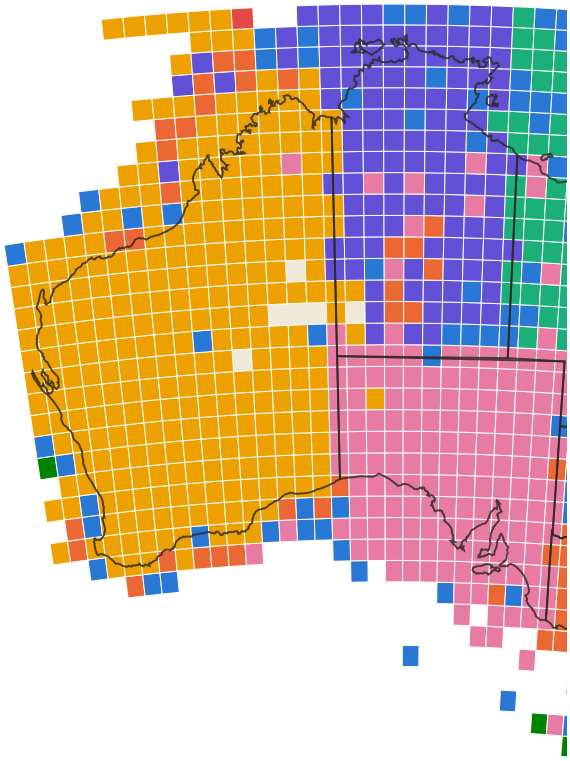


Records per 1° square

under 100 | 100–1k | 1k–5k | 5k–20k | 20k–100k | 100k +

Whose drawer it ended up in

Museum holding the most records from each square



Leading museum (click to highlight)

AM | MV | QM | WAM | SAM | TMAG | MAGNT | QVMAG

The maps show the 4.9 million records whose coordinates fall inside the mapped area. Squares with fewer than 10 records are left out.

NO. 06 · WHEN SPECIMENS WERE COLLECTED

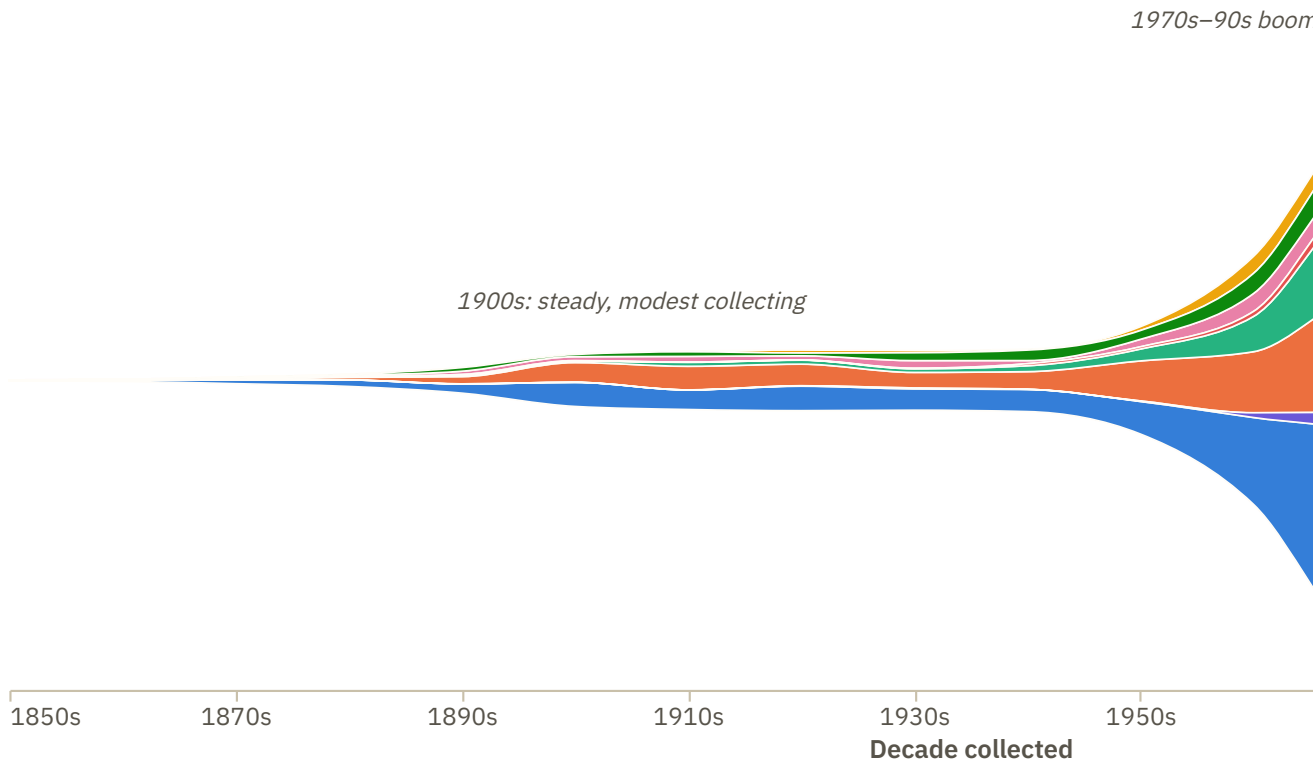
Most specimens were collected after 1960

Specimens from the 1800s are prized, but they're rare. The stream below shows when each dated record was collected. Collecting stayed modest until the

1950s, then took off as governments funded **biological surveys** of the reef, the outback and new national parks. The peak came in the **1990s**, with 815,000 records in a single decade. Collecting has slowed everywhere since then. So far in the 2020s, Hobart's museum has added more dated specimens than any other.

When the specimens were collected

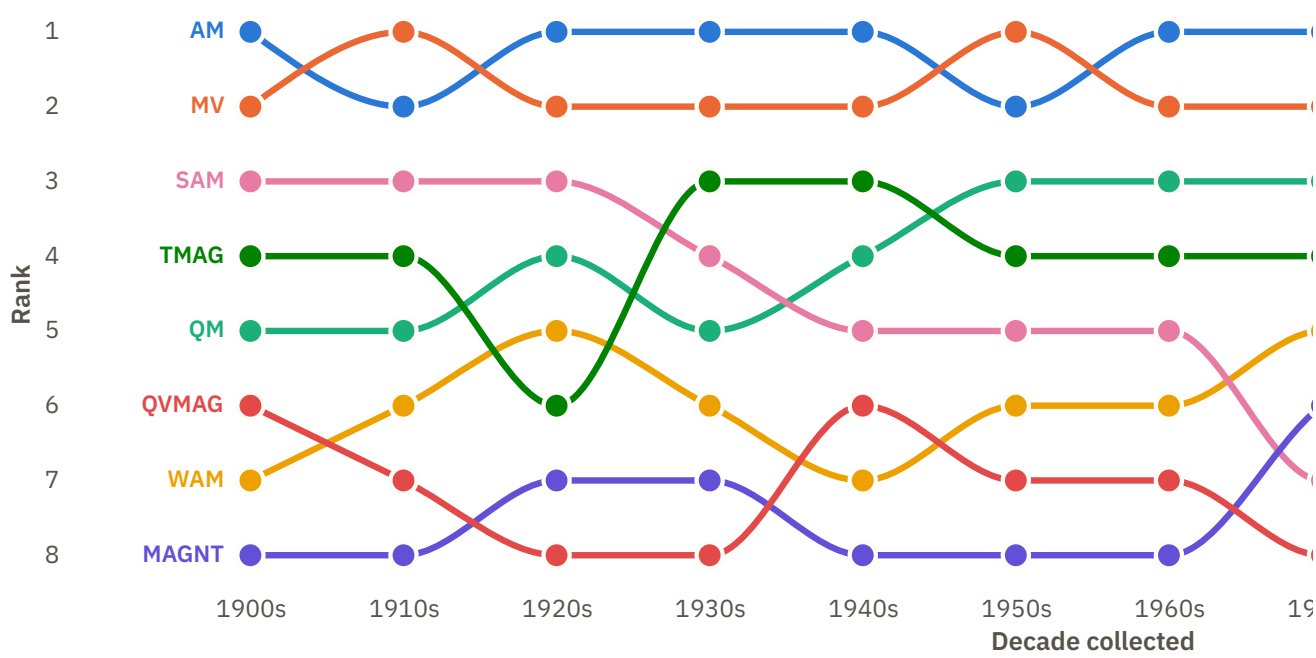
Records by decade, stacked around a centre line



1.6 million records (27%) have no collection date and aren't shown. The 2020s column covers 2020 to October 2026 only.

Who collected the most, decade by decade

Rank by number of dated records (1 = most)



For nine decades, Sydney and Melbourne swapped first and second place. Then the order shuffled: Queensland took the lead in the 2000s, and in the 2010s Tasmania and Western Australia climbed to second and third, overtaking much larger museums.

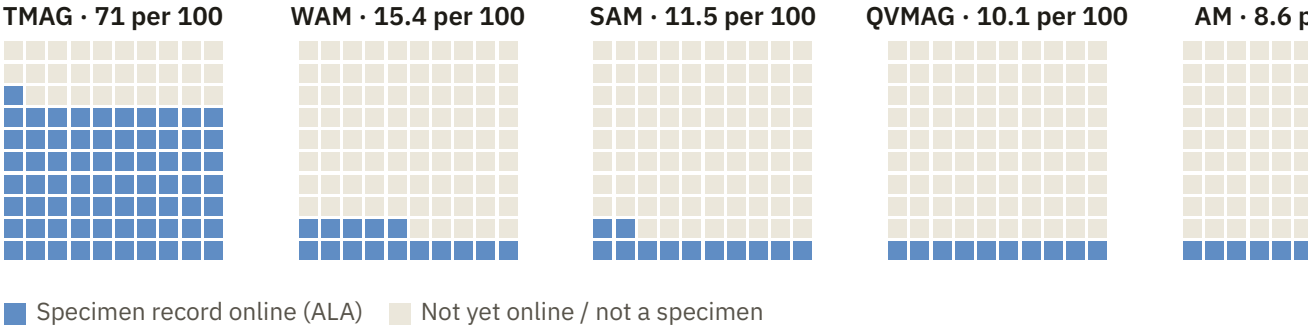
NO. 07 · WHAT YOU CAN SEE ONLINE

Only a small part of each collection is online

For every 100 objects in the big museums' stores, fewer than ten have a searchable specimen record online. Hobart is the outlier: its herbarium sheets were digitised early, so there are 71 online records for every 100 items it holds.

How much of each collection is online

Online specimen records for every 100 objects held (each square = 1)



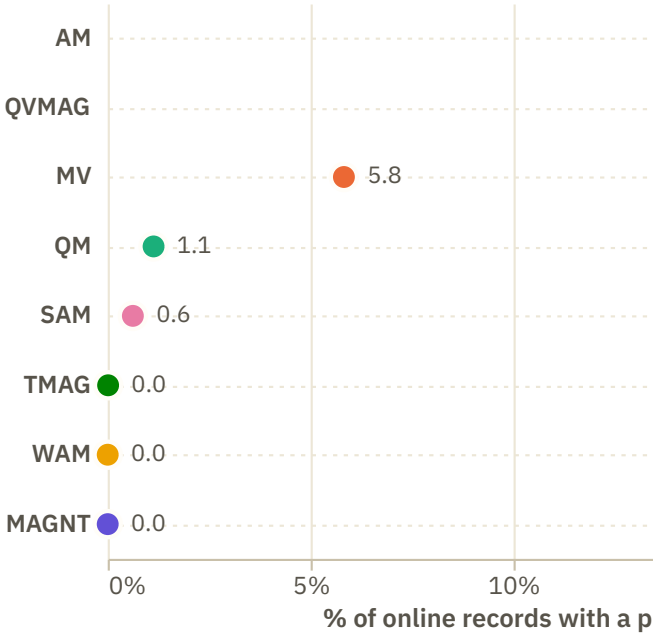
This is a rough guide, not an exact percentage. Collection totals include artworks, cultural objects and history items that the ALA doesn't cover, and one record can describe a jar holding many specimens. Darwin is left out because it publishes no total.

A record is a start, but it's only a line of text. Only **400,000** of the 6.1 million online records (6.6%) include a photograph. The Australian Museum and Launceston's QVMAG are far ahead, with photos on about one record in seven. Three museums have almost no photos online.

Australia's natural heritage is well looked after in physical drawers. Opening those drawers to everyone — online, with pictures — is the job of the next decade.

Records you can actually look at

% of online records with at least one photo



About this visualisation

Data sources

1. **Museum collection totals** from each museum's own publications: Australian Museum (also department figures); Museums Victoria Annual Report 2023–24; Queensland Museum Annual Report 2023–24; South Australian Museum; TMAG (CAMD profile); MAGNT Annual Report 2021–22.